Number System & HCF-LCM — Full Solutions

Step-by-step solutions for all 30 Practice Questions

Q1. Find the HCF of 84 and 126.

Solution: $84 = 2^2\times3\times7$, $126 = 2\times3^2\times7$.
HCF = $2\times3\times7 = 42$.
(Euclid check: $126 = 1\times84+42$; $84 = 2\times42+0 \rightarrow$ HCF=42)
Answer: 42

Q2. Find the LCM of 15, 20 and 25.

Solution: $15 = 3\times5$, $20 = 2^2\times5$, $25 = 5^2$.
LCM = $2^2\times3\times5^2 = 4\times3\times25 = 300$.
Answer: 300

Q3. The HCF and LCM of two numbers are 6 and 210. If one number is 30, find the other.

Solution: Product of numbers = HCF × LCM = $6\times210 = 1260$.
Other number = $1260/30 = 42$.
Answer: 42

Q4. Find the greatest number that divides 285 and 1249 leaving remainders 9 and 7 respectively.

Solution: Required number divides $(285–9)=276$ and $(1249–7)=1242$ exactly.
$276 = 2^2\times3\times23$, $1242 = 2\times3^3\times23$.
HCF(276,1242) = $2\times3\times23 = 138$.
Answer: 138

Q5. Find the least number which when divided by 5, 6, 7 and 8 leaves remainder 3 in each case.

Solution: LCM(5,6,7,8): 5, $6=2\times3$, 7, $8=2^3$.
LCM = $2^3\times3\times5\times7 = 840$.
Required number = $840+3 = 843$.
Answer: 843

Q6. Three bells toll at intervals of 9, 12 and 15 minutes. If they toll together at 8:00 AM, when will they next toll together?

Solution: LCM(9,12,15): $9=3^2$, $12=2^2\times3$, $15=3\times5$.
LCM = $2^2\times3^2\times5 = 180$ minutes = 3 hours.
Next toll together = 8:00 AM + 3 hrs = 11:00 AM.
Answer: 11:00 AM

Q7. Find the smallest number which when increased by 5 is divisible by 24, 32 and 36.

Solution: LCM(24,32,36): $24=2^3\times3$, $32=2\blacksquare$, $36=2^2\times3^2$.
LCM = $2\blacksquare\times3^2 = 288$.
Number + 5 = 288 $\rightarrow$ Number = $288–5 = 283$.
Answer: 283

Q8. Find the number of factors of 360.

Solution: $360 = 2^3 \times 3^2 \times 5^1$.
Number of factors = (3+1)(2+1)(1+1) = 4×3×2 = 24.
Answer: 24

Q9. Find the sum of all factors of 60.

Solution: $60 = 2^2 \times 3 \times 5$.
Sum of factors = (1+2+4)(1+3)(1+5) = 7×4×6 = 168.
Answer: 168

Q10. Find the unit digit of 3^57.

Solution: Cyclicity of 3 (cycle length 4): 3,9,7,1.
57 mod 4 = 1 → 1st value in cycle = 3.
Unit digit = 3.
Answer: 3

Q11. Find the unit digit of 24^36 × 17^47.

Solution: Unit digit of 24 is 4; cycle of 4 (length 2): 4,6.
36 mod 2 = 0 → use 2nd value = 6. So unit digit of 24^36 = 6.
Unit digit of 17 is 7; cycle of 7 (length 4): 7,9,3,1.
47 mod 4 = 3 → 3rd value = 3. So unit digit of 17^47 = 3.
Product unit digit = (6×3) mod 10 = 18 mod 10 = 8.
Answer: 8

Q12. Find the number of trailing zeros in 50!.

Solution: Zeros = ■50/5■ + ■50/25■ + ■50/125■ = 10 + 2 + 0 = 12.
Answer: 12

Q13. Find the number of trailing zeros in 125!.

Solution: Zeros = ■125/5■+■125/25■+■125/125■+■125/625■ = 25+5+1+0 = 31.
Answer: 31

Q14. How many numbers between 1 and 100 are divisible by both 3 and 5?

Solution: Divisible by both 3 and 5 ⇒ divisible by 15.
■100/15■ = 6.
Answer: 6

Q15. Two numbers are in the ratio 4:5 and their HCF is 6. Find their LCM.

Solution: Numbers = 4×6=24 and 5×6=30.
LCM = (24×30)/HCF = 720/6 = 120.
(Or directly: LCM = 4×5×6 = 120 since 4,5 are co-prime.)
Answer: 120

Q16. Find the HCF of 2/3, 4/9 and 6/27.

Solution: HCF of fractions = HCF(numerators)/LCM(denominators).
Numerators: 2,4,6 → HCF = 2.
Denominators: 3,9,27 → LCM = 27.
HCF = 2/27.
Answer: 2/27

Q17. Find the LCM of 3/4, 5/6 and 7/12.

Solution: LCM of fractions = LCM(numerators)/HCF(denominators).
Numerators: 3,5,7 → LCM = 105.
Denominators: 4,6,12 → HCF = 2.
LCM = 105/2.
Answer: 105/2

Q18. What is the largest 4-digit number exactly divisible by 88?

Solution: 9999 ÷ 88 = 113.62 → take 113.
113 × 88 = 9944.
Answer: 9944

Q19. What is the smallest 5-digit number exactly divisible by 476?

Solution: Smallest 5-digit number = 10000.
10000 ÷ 476 = 21.00... → next integer = 22.
22 × 476 = 10472.
Answer: 10472

Q20. Find the remainder when 2^100 is divided by 7.

Solution: $2^3 = 8 \equiv 1 \pmod 7$, so cycle length = 3.
100 mod 3 = 1 → 2^100 ≡ 2^1 ≡ 2 (mod 7).
Answer: 2

Q21. Find the remainder when 7^105 is divided by 5.

Solution: 7 ≡ 2 (mod 5). Cycle of 2 mod 5 (length 4): 2,4,3,1.
105 mod 4 = 1 → 1st value = 2.
Remainder = 2.
Answer: 2

Q22. How many numbers from 1 to 200 are co-prime to 200?

Solution: $200 = 2^3 \times 5^2$.
ϕ(200) = 200×(1–1/2)×(1–1/5) = 200×(1/2)×(4/5) = 80.
Answer: 80

Q23. A number when divided by 296 leaves remainder 75. What remainder is obtained when the same number is divided by 37?

Solution: 296 = 8×37, so 296 is a multiple of 37.
Remainder on dividing by 37 = 75 mod 37 = 75–74 = 1.
Answer: 1

Q24. Find the sum of first 40 natural numbers.

Solution: Sum = n(n+1)/2 = 40×41/2 = 820.
Answer: 820

Q25. Find the sum of squares of first 15 natural numbers.

Solution: Sum = n(n+1)(2n+1)/6 = 15×16×31/6 = 7440/6 = 1240.
Answer: 1240

Q26. If $N = 2^3 \times 3^2 \times 5$, find the number of even factors of N.

Solution: Total factors = (3+1)(2+1)(1+1) = 24.
Odd factors (no factor of 2) = (2+1)(1+1) = 6.
Even factors = 24 – 6 = 18.
Answer: 18

Q27. Find the HCF of 96, 144 and 192.

Solution: 96 = 2■×3, 144 = 2■×3^2, 192 = 2■×3.
HCF = 2■×3 = 48.
Answer: 48

Q28. The product of two numbers is 1320 and their HCF is 6. Find their LCM.

Solution: Product = HCF × LCM → LCM = 1320/6 = 220.
Answer: 220

Q29. Find the least number which is a perfect square and is divisible by 16, 18 and 45.

Solution: LCM(16,18,45): 16=2■, 18=2×3^2, 45=3^2×5.
LCM = 2■×3^2×5 = 720 = 2■×3^2×5^1.
Power of 5 is odd, so multiply by 5 to make all powers even.
720×5 = 3600 = 2■×3^2×5^2 = 60^2.
Answer: 3600

Q30. Check whether 1001 is divisible by 11. If yes, find the quotient.

Solution: Divisibility rule of 11: alternating digit sum (1–0+0–1) = 0, which is divisible by 11. So yes.
1001 ÷ 11 = 91.
Answer: Yes, quotient = 91